

EMX Layout Design: On-Chip Inductor

Inductance (L) and Quality Factor (Q) Extraction

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1. Objective

Design a spiral inductor and characterize its electrical performance using EMX electromagnetic simulation. The inductance (L) and quality factor (Q) are extracted versus frequency from the simulated Y/S-parameters, and their values are reported at 11.7 GHz, the target operating frequency for the LNA design.

2. Inductor Layout

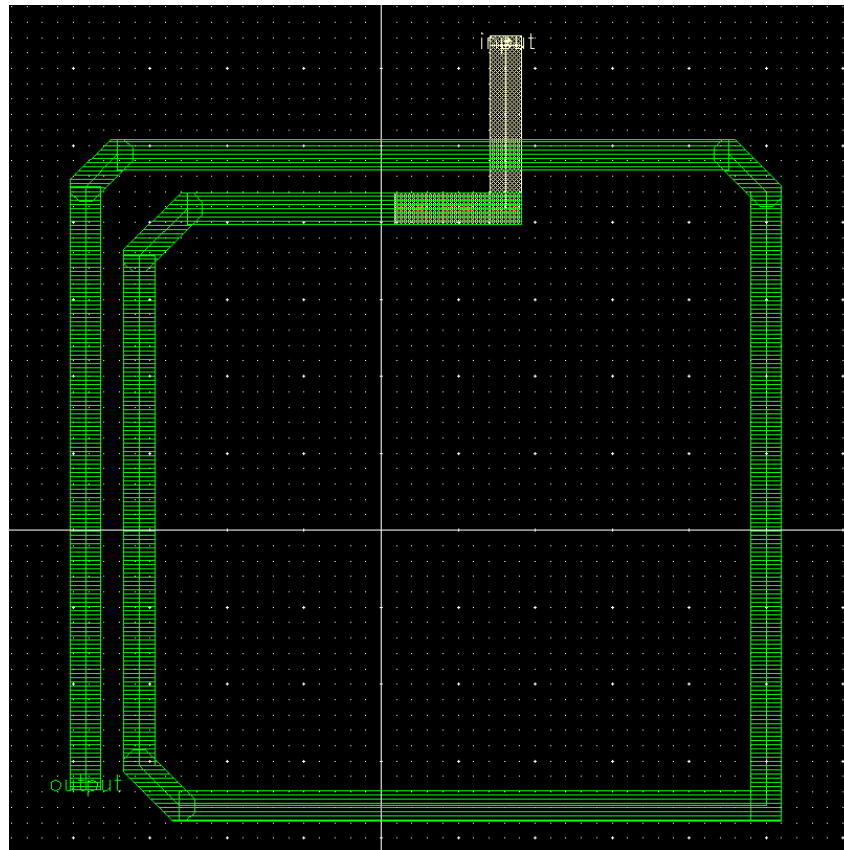


Figure 1. EMX layout view of the spiral inductor: top-down metal geometry showing the spiral trace, port terminals (input/output).

3. Simulation Results

The inductor was simulated in EMX over a swept frequency range, and the two-port S-parameters were converted to an equivalent one-port inductance and quality factor. Results are shown below.

3.1 Inductance vs. Frequency

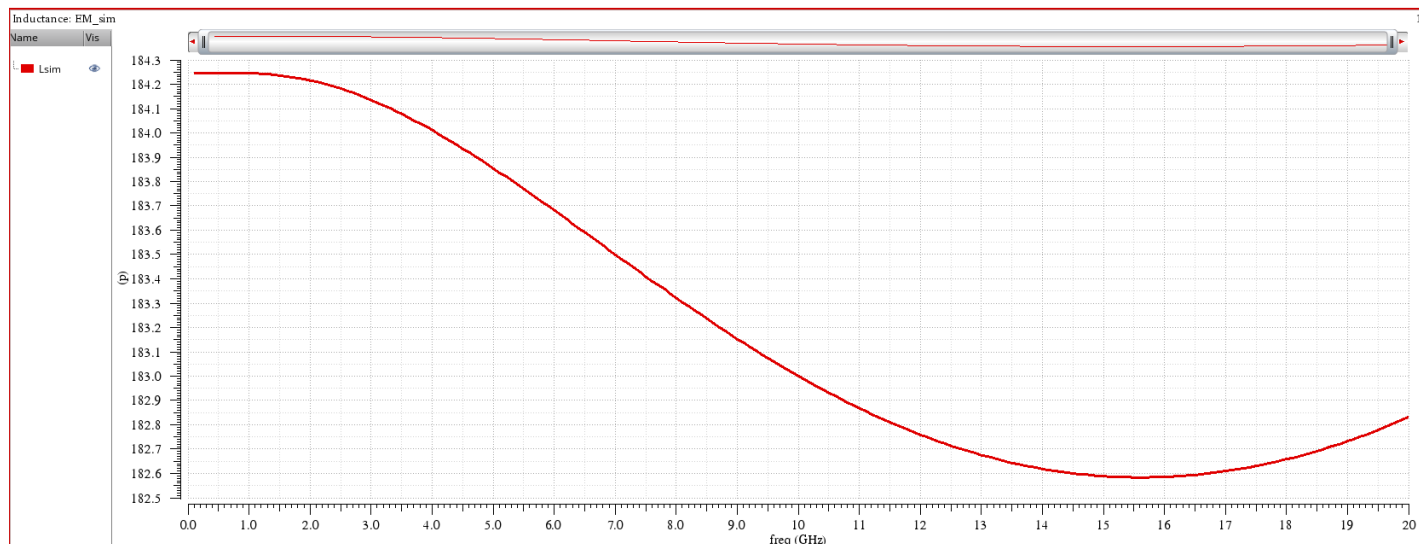


Figure 2. Extracted inductance L (pH, y-axis) vs. frequency (GHz, x-axis), from EMX S-parameter simulation of the layout in Figure 1.

Comment: L stays relatively flat across the swept band and rises sharply as the inductor approaches self-resonance — expected spiral-inductor behavior.

3.2 Quality Factor vs. Frequency

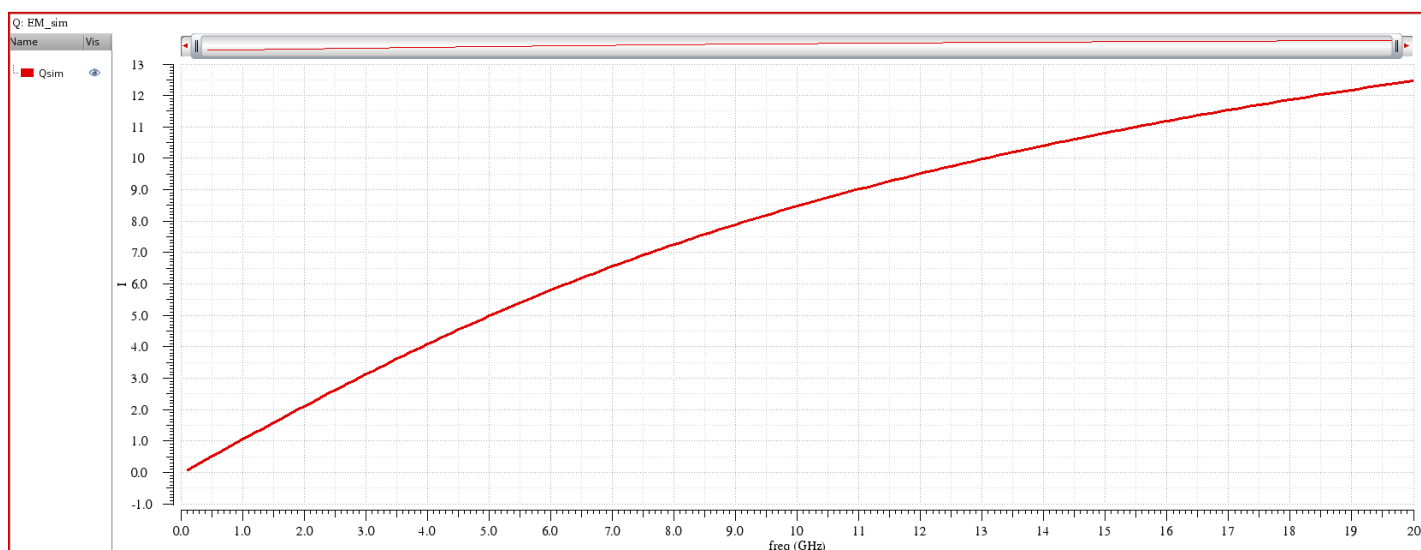


Figure 3. Extracted quality factor Q (dimensionless, y-axis) vs. frequency (GHz, x-axis), from the same EMX simulation as Figure 2.

Comment: Q increases with frequency, peaks, then drops toward zero near self-resonance as losses and reactive coupling dominate — again the standard trend for on-chip spirals.

4. Value at Design Frequency (11.7 GHz, for LNA design)

Markers were placed on the L and Q curves at 11.7 GHz, the intended LNA operating frequency, to read off the design values used downstream.

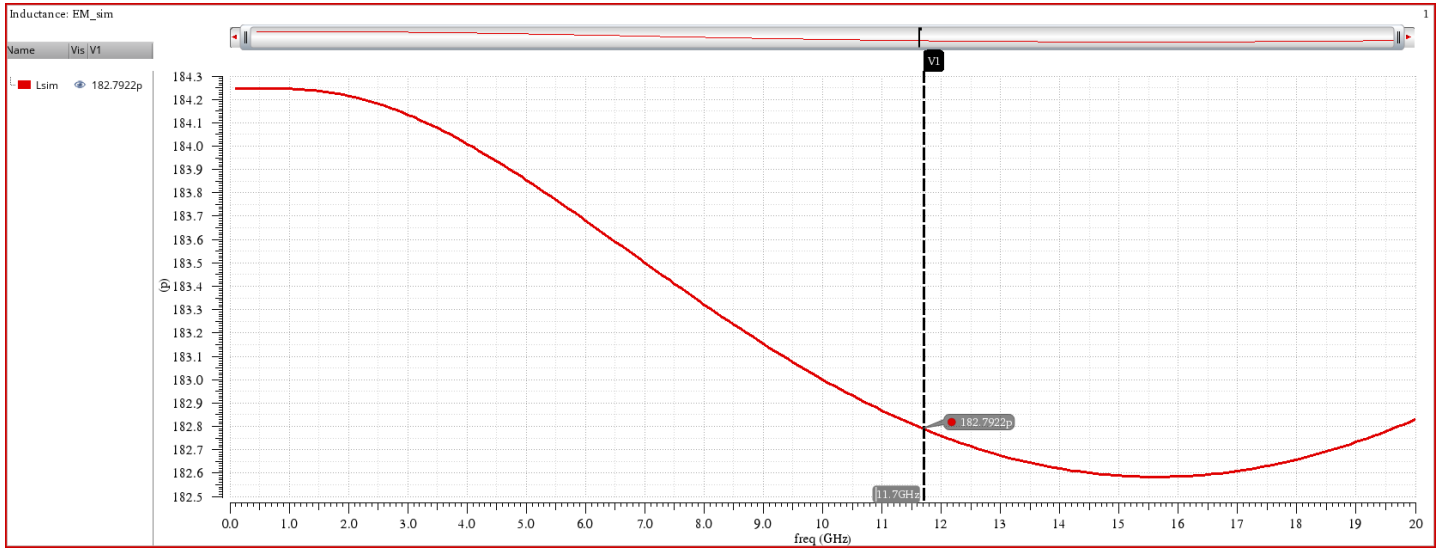


Figure 4. Zoomed L vs. frequency (GHz) plot with a marker at 11.7 GHz reading $L = 182.79$ pH.

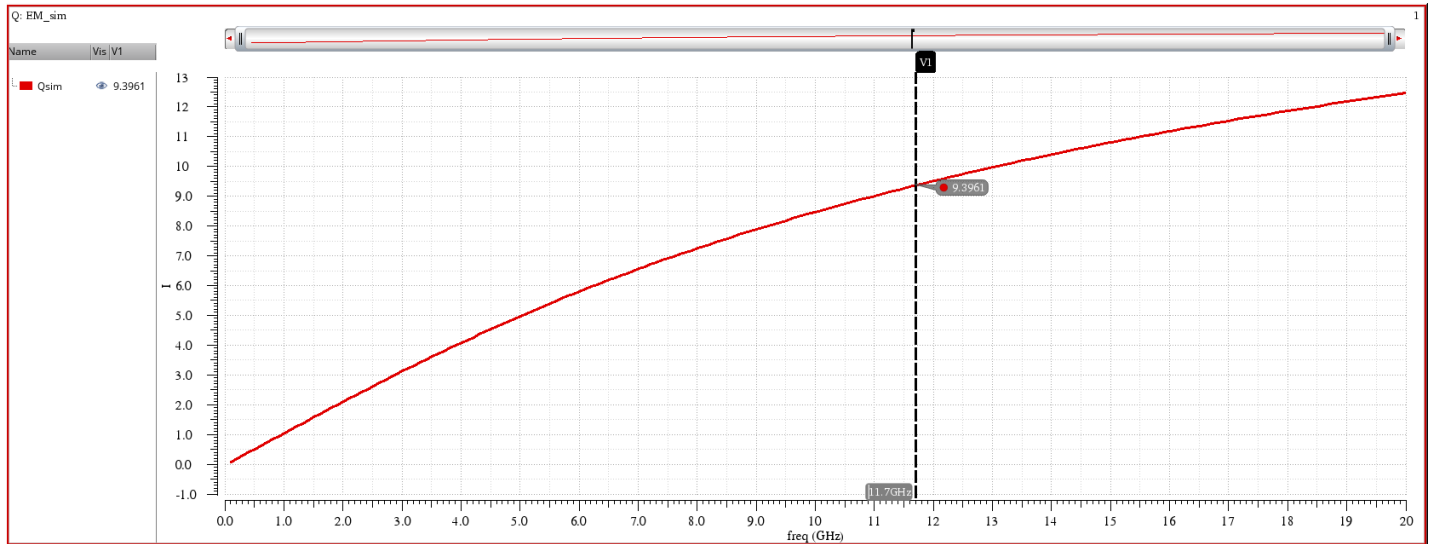


Figure 5. Zoomed Q vs. frequency (GHz) plot with a marker at 11.7 GHz reading $Q = 9.3961$.

Parameter	Value	Frequency
Inductance, L	182.79 pH	11.7 GHz
Quality Factor, Q	9.3961	11.7 GHz

Comment: At 11.7 GHz the inductor delivers $L \approx 183$ pH with $Q \approx 9.4$, a reasonable Q for an on-chip spiral at this frequency — values ready to be used directly in the LNA design.

5. Conclusion

The designed spiral inductor was successfully simulated in EMX and its L and Q were extracted vs. frequency. At the LNA design frequency of 11.7 GHz, the inductor provides $L = 182.79$ pH and $Q = 9.3961$, meeting the requirement for use in the LNA design.